

MLA0103-Artificial Intelligence with Expert Systems
Assessment Tool

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Heart Disease Prediction Using Machine Learning

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Problem Statement

Heart disease is one of the leading causes of death worldwide. Many patients do not recognize the early symptoms or delay consulting a doctor, which can lead to severe health complications. Traditional diagnosis requires multiple medical tests and expert analysis, making the process time-consuming and expensive. There is a need for an intelligent system that can assist healthcare professionals by predicting the likelihood of heart disease using patient health data. To address this problem, a **Heart Disease Prediction System Using Machine Learning** is developed. The system analyzes medical parameters such as age, blood pressure, cholesterol level, heart rate, chest pain type, and other clinical features to predict whether a patient is at risk of heart disease. This enables early detection, faster decision-making, and improved patient care.

Agent Name: Heart Disease Prediction System

Type of Agent: Machine Learning-Based Predictive Healthcare Agent

Google Dialogflow:

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from historical data and make predictions without being explicitly programmed. In this project, supervised machine learning algorithms are trained using a heart disease dataset.

Environment:

The Heart Disease Prediction System operates in a software environment using **Python** with libraries such as **NumPy**, **Pandas**, **Scikit-learn**, **Matplotlib**, and **Flask** (optional for web deployment). The system can be deployed as a desktop application, web application, or integrated into hospital management systems for real-time prediction.

Inputs

The Heart Disease Prediction System accepts patient medical data such as:

- Age
- Gender
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol Level
- Fasting Blood Sugar
- Resting ECG Results
- Maximum Heart Rate
- Exercise-Induced Angina
- ST Depression (Oldpeak)
- Number of Major Blood Vessels
- Thalassemia

Outputs

The Heart Disease Prediction System provides prediction results such as:

- Heart Disease Detected (High Risk)
- No Heart Disease Detected (Low Risk)
- Heart Disease Risk Prediction
- Prediction Confidence Score
- Patient Health Risk Assessment
- Clinical Decision Support for Doctors

Objectives:

The main objectives of the **Heart Disease Prediction Using Machine Learning** system are:

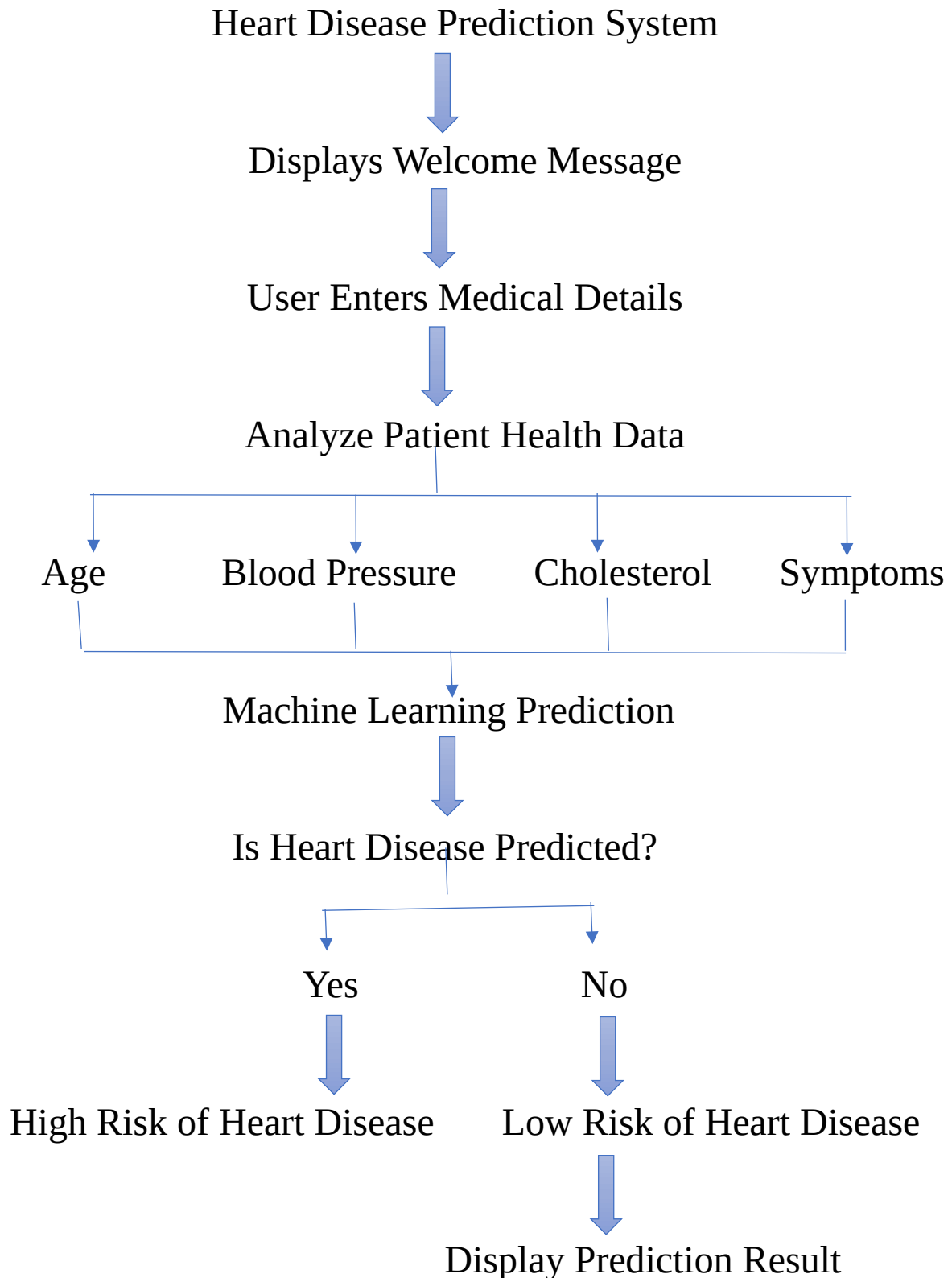
- To predict the risk of heart disease using patient medical data.
- To assist doctors in making faster and more accurate diagnoses.
- To enable early detection of heart disease and improve patient care.
- To reduce the time required for manual health risk assessment.
- To provide accurate and reliable heart disease predictions using machine learning algorithms.

Knowledge Base

The Heart Disease Prediction System stores and utilizes information related to:

- Patient Medical Records
- Age and Gender Details
- Blood Pressure Information
- Cholesterol Levels
- Blood Sugar Levels
- ECG (Electrocardiogram) Results
- Chest Pain Type
- Maximum Heart Rate
- Exercise-Induced Angina
- Thalassemia Information
- Heart Disease Dataset
- Machine Learning Prediction Models

Conversation Flow Diagram:



Selected Search Strategy

Machine Learning-Based Classification

For this project, Machine Learning-Based Classification is selected because the system uses trained machine learning algorithms to analyze patient medical data and predict the likelihood of heart disease. The model learns patterns from historical healthcare datasets and classifies patients as either high risk or low risk based on their health parameters.

Why Heuristic Search?

Machine learning classification is suitable for heart disease prediction because it:

- Quickly predicts the risk of heart disease.
- Improves prediction accuracy using historical medical data.
- Assists doctors in early diagnosis and decision-making.
- Efficiently analyzes multiple patient health parameters.
- Reduces manual effort and provides fast prediction results.
- Enhances healthcare services through data-driven analysis.

Since the system uses trained **Machine Learning algorithms** such as **Logistic Regression, Decision Tree, Random Forest**, or **Support Vector Machine (SVM)**, classification is naturally integrated into the heart disease prediction process.

Advantages of the Developed Chatbot

- Provides early prediction of heart disease risk.
- Assists doctors in making faster and more accurate diagnoses.
- Reduces the time required for manual health assessment.
- Improves the accuracy of disease prediction using machine learning.
- Supports timely medical intervention and better patient care.
- Analyzes multiple health parameters efficiently.
- Enhances healthcare decision-making through data-driven insights.

Conclusion:

The Heart Disease Prediction Using Machine Learning system was successfully tested using patient health data from a heart disease dataset. The machine learning model accurately analyzed medical parameters and predicted the likelihood of heart disease with high accuracy. The evaluation demonstrated reliable performance, fast prediction time, and effective classification of patients into high-risk and low-risk categories. With future enhancements such as real-time hospital database integration, wearable health sensor support, and improved machine learning models, the system can become a valuable decision-support tool for doctors and healthcare professionals.

